

Rutwik Dakhore

AI Engineer | Machine Learning & MLOps

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with a PG Diploma in Artificial Intelligence (CDAC) and a B.E. in Electronics & Telecommunication (CGPA 9.7). Builds end-to-end machine learning systems across NLP, computer vision, and time-series forecasting, from data preprocessing and model training to validation and deployment in Streamlit apps. Prior experience in CI/CD automation with Jenkins at Amdocs, with working knowledge of Docker and MLOps tooling (MLflow, Airflow, DVC).

TECHNICAL SKILLS

Programming:	Python, Java, SQL, C, C++
AI & Machine Learning:	TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, NLTK
Generative AI:	Transformers, Hugging Face, LangChain, LangGraph, RAG, Prompt Engineering
Data & Analytics:	Pandas, NumPy, Exploratory Data Analysis (EDA), Feature Engineering
Big Data & Streaming:	Apache Spark (PySpark), Apache Kafka
MLOps & Automation:	Docker, Kubernetes, Jenkins, MLflow, Airflow, DVC
Visualization:	Matplotlib, Seaborn, Power BI
Databases & Tools:	MySQL, Git

PROJECTS

RAG-Based Document Question Answering System **Aug 2026 – Sept 2026**
Python, FastAPI, Next.js, FAISS, Ollama (Llama 3), PyMuPDF, Tailwind CSS GitHub Repository

- Developed a full-stack **Retrieval-Augmented Generation (RAG)** application that lets users upload PDF, DOCX, and TXT documents and query them with natural-language questions.
- Built the ingestion pipeline with PyMuPDF and python-docx for parsing, text chunking, and **semantic embeddings** (all-MiniLM-L6-v2), indexed in a **FAISS vector database** for top-k similarity search.
- Integrated **Ollama** for local LLM inference (Llama 3), generating context-grounded answers with source and page citations, enabling privacy-focused, offline document processing.
- Exposed ingestion, retrieval, and generation through **FastAPI REST endpoints**; built a responsive **Next.js/TypeScript** interface with Tailwind CSS for document management, index rebuilding, retrieved-context inspection, and relevance scores.

Veritas AI – Multimodal Fake News Detection **Dec 2025 – Jan 2026**
Python, TensorFlow, Scikit-learn, NLP, Computer Vision, Pandas, NumPy, Streamlit GitHub Repository

- Built a multimodal fake news detection system combining NLP and computer vision, covering data preprocessing, feature extraction, model training, and validation.
- Designed independent text and image pipelines for feature extraction and classification, fused through adjustable weights.
- Achieved **85.7% accuracy** and **0.85 macro-F1** on a 4,752-sample validation set, against a 61% majority-class baseline.
- Deployed a Streamlit app returning a verdict, confidence score, and per-modality probabilities, with a fallback for missing or invalid images.

Customer Segmentation for Online Retail
Python, Pandas, Scikit-learn, Matplotlib, Seaborn

Mar 2023 – May 2023
GitHub Repository

- Performed exploratory data analysis, preprocessing, and clustering (K-means and hierarchical) to segment customers by purchasing behavior.
- Identified four key customer segments and visualized their patterns to support targeted marketing strategies.

Bike Sharing Demand Forecasting
Python, Scikit-learn, Keras, LSTM, Pandas

Jan 2023 – Mar 2023
GitHub Repository

- Built and validated demand forecasting models using linear regression, decision trees, and LSTM, with feature engineering and metric-based evaluation.
- Modeled time-series patterns to predict daily bike demand for ride-sharing systems.

Mobile Price Range Prediction
Python, Scikit-learn, XGBoost, Pandas

Oct 2022 – Dec 2022
GitHub Repository

- Developed and validated classification models using Random Forest and XGBoost, achieving **~85% accuracy** on unseen data.
- Performed preprocessing, feature engineering, hyperparameter tuning, and model evaluation to improve predictive performance.

PROFESSIONAL EXPERIENCE

Software Engineering Associate
AMDOCS

Oct 2021 – May 2022

- Designed and automated CI/CD pipelines using Jenkins, reducing deployment time by **40%**.
- Integrated non-production and production configuration maps, improving configuration management.
- Built Jenkins jobs to encrypt sensitive credentials in logs, strengthening compliance and security.
- Collaborated with cross-functional teams and maintained technical documentation for deployment workflows, CI/CD processes, and configuration management.

EDUCATION

PG Diploma in Artificial Intelligence (PG-DAI)
Centre for Development of Advanced Computing (CDAC), Guwahati

Completed Jan 2026
80.67% – Grade A

Bachelor of Engineering, Electronics & Telecommunication
D.M.I.E.T.R., Wardha

Graduated 2021
CGPA: 9.7